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ORIGINAL ARTICLE – RADIATION ONCOLOGY

Radiation Therapy is Associated with Improved Outcomes in Merkel Cell Carcinoma

Tobin Strom, MD[1], Michael Carr, MS[2], Jonathan S. Zager, MD[3,4], Arash Naghavi, MD, MS[1], Franz O. Smith, MD[3,4], C. Wayne Cruse, MD[3,4], Jane L. Messina, MD[3,5], Jeffery Russell, MD, PhD[6], Nikhil G. Rao, MD[1], William Fulp, MS[7], Sungjune Kim, MD, PhD[1], Javier F. Torres-Roca, MD[1], Tapan A. Padhya, MD[6], Vernon K. Sondak, MD[3,4], Andy M. Trotti, MD[1], Louis B. Harrison, MD, FASTRO[1], and Jimmy J. Caudell, MD, PhD[1]

[1]Department of Radiation Oncology, H. Lee Moffitt Cancer Center, Tampa, FL; [2]School of Medicine, University of South Florida Morsani College of Medicine, Tampa, FL; [3]Department of Cutaneous Oncology, Moffitt Cancer Center, Tampa, FL; [4]Departments of Oncologic Sciences and Surgery, University of South Florida Morsani College of Medicine, Tampa, FL; [5]Departments of Pathology & Cell Biology and Dermatology, University of South Florida Morsani College of Medicine, Tampa, FL; [6]Department of Head and Neck and Endocrine Oncology, Moffitt Cancer Center, Tampa, FL; [7]Department of Biostatistics, Moffitt Cancer Center, Tampa, FL

ABSTRACT

Background. Following wide excision of Merkel cell carcinoma (MCC), postoperative radiation therapy (RT) is typically recommended. Controversy remains as to whether RT can be avoided in selected cases, such as those with negative margins. Additionally, there is evidence that RT can influence survival.

Methods. We included 171 patients treated for non-metastatic MCC from 1994 through 2012 at a single institution. Patients without pathologic nodal evaluation (clinical N0 disease) were excluded to reflect modern treatment practice. The endpoints included local control (LC), locoregional control (LRC), disease-free survival (DFS), overall survival (OS), and disease-specific survival (DSS).

Results. Median follow-up was 33 months. Treatment with RT was associated with improved 3-year LC (91.2 vs. 76.9 %, respectively; $p = 0.01$), LRC (79.5 vs. 59.1 %; $p = 0.004$), DFS (57.0 vs. 30.2 %; $p < 0.001$), and OS (73 vs. 66 %; $p = 0.02$), and was associated with improved 3-year DSS among node-positive patients (76.2 vs. 48.1 %; $p = 0.035$), but not node-negative patients (90.1 vs. 80.8 %; $p = 0.79$). On multivariate analysis, RT was associated with improved LC [hazard ratio (HR) 0.18, 95 % confidence interval (CI) 0.07–0.46; $p < 0.001$], LRC (HR 0.28, 95 % CI 0.14–0.56; $p < 0.001$), DFS (HR 0.42, 95 % CI 0.26–0.70; $p = 0.001$), OS (HR 0.53, 95 % CI 0.31–0.93; $p = 0.03$), and DSS (HR 0.42, 95 % CI 0.26–0.70; $p = 0.001$). Patients with negative margins had significant improvements in 3-year LC (90.1 vs. 75.4 %; $p < 0.001$) with RT. Deaths not attributable to MCC were relatively evenly distributed between the RT and no RT groups (28.5 and 29.3 % of patients, respectively).

Conclusions. RT for MCC was associated with improved LRC and survival. RT appeared to be beneficial regardless of margin status.

Louis B. Harrison and Jimmy J. Caudell were designated as co-senior authors on this work.

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L. B. Harrison, MD, FASTRO
e-mail: louis.harrison@moffitt.org

J. J. Caudell, MD, PhD
e-mail: jimmy.caudell@moffitt.org

Merkel cell carcinoma (MCC) is a rare cutaneous malignancy of neuroendocrine origin, which typically occurs in elderly patients in areas of the body with frequent sun exposure, including the head and neck.[1,2] MCC also has a higher incidence in the severely immunosuppressed population, and has recently been found to be associated with infection by the Merkel cell polyomavirus.[3,4] Despite its rarity, MCC is a highly lethal skin cancer with survival rates at 5 years ranging from 23 to 80 %.[5–9] Treatment of non-metastatic MCC typically includes wide excision with 1–2 cm margins.[10–14] Given the high risk of locoregional recurrence following surgery alone, adjuvant radiation therapy (RT) is typically recommended.

A number of retrospective clinical reports and a single prospective study suggest that, much like other small round blue cell tumors, MCC is radiosensitive, and the use of radiotherapy is associated with improvements in local control, locoregional control (LRC), and/or disease-free survival (DFS).[15–24] While no single-institution retrospective series to date has demonstrated an overall survival (OS) benefit, perhaps due to limited numbers, a Surveillance, Epidemiology and End Results (SEER) study including 1665 MCC patients demonstrated an OS benefit with the use of RT.[25] However, as many patients with MCC are elderly and/or immunosuppressed with competing causes of mortality, and the potential selection biases that might favor radiating healthier patients less likely to die of other causes, conclusions about RT and survival have been limited.

With this background in mind, we reviewed our experience with RT for non-metastatic MCC. We tested the hypothesis that RT would improve both LRC and survival, and also evaluated whether RT would benefit patients regardless of margin status.

METHODS

Patients and Histopathologic Analysis

We performed a retrospective, single-institution review of all patients diagnosed with MCC from 1994 through 2012 after obtaining Institutional Review Board approval. Patients who presented with distant metastasis or locally recurrent disease, who had received prior RT to the area, or who had unavailable treatment records, clinical N0 disease without pathologic nodal evaluation, or less than 12 months of follow-up were excluded. In total, 171 patients with non-metastatic MCC were included in the analysis. Pathologic node-negative patients included those with the American Joint Committee on Cancer (AJCC) staging system nodal stage pN0 and overall stage IA/IIA/IIC; patients with pathologic node-positive disease included those with AJCC nodal stage pN1 and overall stage IIIA; and patients with clinical node-positive disease included those with AJCC nodal stage cN1 and overall stage IIIB. Demographic, histopathologic, RT and outcome data were abstracted from the chart.

Primary Tumor Treatment Characteristics

The majority of patients were treated with wide excision ($n = 168$, 98.2 %), defined as a minimum margin of 1–2 cm as clinically assessed by the surgeon, while a minority of patients were treated with excisional biopsy without wide excision ($n = 3$, 1.8 %). Patients with margins >1 mm were considered to have negative margins. All patients who received RT were treated with external beam RT to the primary site tumor bed, represented by the scar plus a margin of 3–5 cm.

Regional Nodal Treatment Characteristics

All patients underwent surgical evaluation of their regional lymph nodes with either sentinel lymph node biopsy for clinically negative lymph nodes or therapeutic lymph node dissection for clinically palpable lymph nodes. Sentinel lymph nodes were completely embedded; a single hematoxylin and eosin (H&E) stain and anti-AE1/3 cytokeratin immunohistochemical staining was performed on each tumor block. Sentinel nodes were positive if cytokeratin-positive cells with paranuclear dot-like staining or strong membranous staining were present singly or in clumps; these were not always identified on corresponding H&E stain.

Overall, 150 patients (87.7 %) underwent sentinel lymph node biopsy, with a positive node found in 52 patients (34.7 %). Of patients with a positive sentinel lymph node, 30 (57.7 %) had a completion lymphadenectomy. Adjuvant RT was delivered after a positive sentinel lymph node biopsy and clinical lymph node dissection in 24 of 30 cases (80 %), and after positive sentinel lymph node biopsy alone in 19 of 22 patients (86.4 %). Twenty-one patients (12.3 %) underwent a therapeutic lymphadenectomy for clinically palpable lymph nodes, 17 (81.0 %) of which were pathologically involved, and adjuvant RT was delivered in 13 of 17 cases (76.5 %).

The regional lymphatics were typically included in the radiation treatment field if the lymph nodes were clinically or pathologically positive for malignancy. Forty-six of 130 (35.4 %) patients received RT at the primary institution, while the remaining 84 (64.6 %) patients were treated with RT at outside radiation oncology centers.

Endpoint and Statistical Analysis

Outcomes of interest were local control, defined as freedom from disease recurrence within 5 cm of the primary tumor; LRC, defined as freedom from disease at the primary tumor bed, in-transit region and regional lymph nodes; DFS, defined as time to disease recurrence anywhere, or death from any cause; OS, defined as time to death from any cause; and disease-specific survival (DSS), defined as death from MCC.

Statistical analysis was performed using SPSS® version 22.0 (IBM Corporation, Armonk, NY, USA). Actuarial rates of local control, LRC, DFS, OS, and DSS were calculated using the Kaplan–Meier method, and differences in rates based on individual variables were assessed using the log-rank test. All clinical, histopathologic, and treatment variables were included in the Cox multivariate analysis models. Continuous variables were split using clinically meaningful cut-points. All tests were two-sided and an α (type I) error ≤ 0.05 was considered to be statistically significant.

RESULTS

Clinicopathologic and Treatment Characteristics

Clinicopathologic and treatment variables among patients treated with and without RT are shown in Appendix Table 1. Of 171 patients included in the study, the median age was 74 years (range 31–96 years), median pathologic tumor size was 1.5 cm (range 0.2–8.0), and the majority of patients were male (72.5 %), with margin-negative (88.3 %) and node-negative (59.1 %) MCC. The median prescribed RT dose was 5000 cGy (range 4000–6600 cGy) in 25 daily fractions (range 15–33) at 200 cGy per fraction (range 180–325 cGy). Among all clinicopathologic and treatment variables, no significant differences between patients treated with and without RT were observed.

Local and Locoregional Control Outcomes

The median follow-up of surviving patients was 33 months (range 12–190). In total, 20 of 171 (11.7 %) patients failed locally, and the median time to local failure was 9 months (range 1–37). Similarly, 27 of 171 (15.8 %) patients experienced a regional recurrence, with a median time to regional failure of 9 months (range 1–42 months). Variables associated with local control and LRC among all patients are shown in Table 2. Among the 41 patients who did not receive RT, nine (22.0 %) and 15 (36.6 %) patients developed a local and locoregional recurrence, respectively, compared with 11 (8.5 %) and 25 (23.9 %) of 130 irradiated patients, respectively. Treatment with RT was associated with improved 3-year local control (Fig. 1a; 91.2 % for RT vs. 76.9 % for no RT; $p = 0.01$) and 3-year LRC (Fig. 1b; 79.5 and 59.1 %, respectively; $p = 0.004$). On multivariate analysis, RT was independently associated with improved local control (Table 1) [hazard ratio (HR) 0.18, 95 % confidence interval (CI) 0.07–0.46; $p < 0.001$] and LRC (HR 0.28, 95 % CI 0.14–0.56; $p < 0.001$).

On subgroup analysis, treatment with RT was associated with improved local control and LRC among both patients who had positive surgical margins and those with negative margins. The largest difference in local control between radiated and non-radiated patients was seen for patients with positive margins. Among 16 patients with positive margins, 2 of 2 (100 %) patients who were not irradiated failed locally compared with 2 of 12 (14.3 %) irradiated patients, with 3-year local control rates of 0 and 85.1 %,

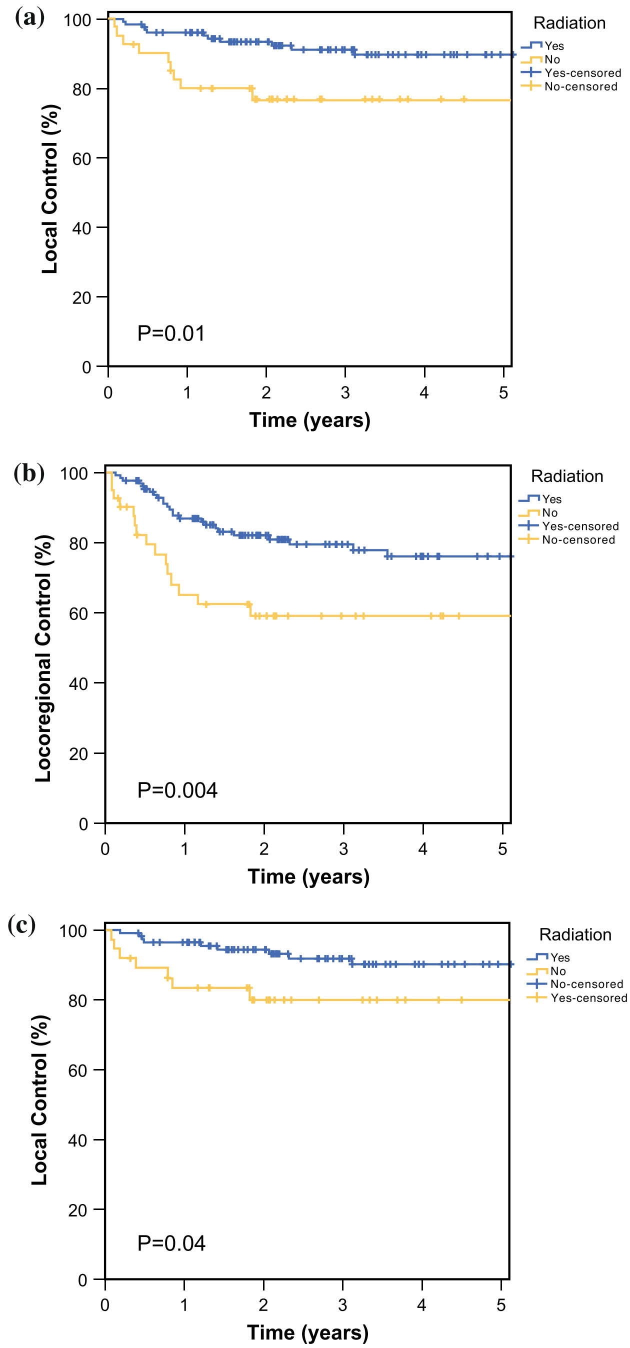

FIG. 1 Kaplan–Meier plots for patients with Merkel cell carcinoma treated with or without radiation therapy demonstrating **a** local control and **b** locoregional control among all patients ($n = 171$), and **c** local control for patients with negative margins ($n = 151$)

TABLE 1 Variables associated with local control and locoregional control among all patients with Merkel cell carcinoma

Variable		Local control				
		KM 3-year	Cox UV analysis		Cox MV analysis	
			HR (95 % CI)	p value	HR (95 % CI)	p value
Lymphovascular invasion	No	89.9				
	Yes	69.5	3.05 (1.11–8.42)	**0.03**	4.51 (1.49–13.61)	**0.008**
Nodal stage	Pathologic N0	94.7	Reference		Reference	
	Pathologic N1	79.1	4.25 (1.45–12.45)	**0.008**	7.40 (2.40–22.81)	**<0.001**
	Clinical N1	71.7	7.32 (2.12–25.33)	**0.002**	9.24 (2.50–34.14)	**0.001**
Radiation	No	76.9				
	Yes	91.2	0.34 (0.14–0.82)	**0.02**	0.18 (0.07–0.46)	**<0.001**
Variable		**Locoregional control**				
		KM 3-year	Cox UV analysis		Cox MV analysis	
			HR (95 % CI)	p value	HR (95 % CI)	p value
Sex	Male	69.4				
	Female	88.7	0.34 (0.13–0.88)	**0.03**	0.34 (0.13–0.88)	**0.03**
Nodal stage	Pathologic N0	81.0				
	Pathologic N1	68.1	1.66 (0.84–3.29)	0.15	2.04 (0.99–4.20)	0.053
	Clinical N1	60.1	2.57 (1.07–6.17)	**0.03**	3.29 (1.33–8.12)	**0.01**
Radiation	No	59.1				
	Yes	79.5	0.40 (0.21–0.77)	**0.006**	0.28 (0.14–0.56)	**<0.001**

CI confidence interval, *HR* hazard ratio, *KM* Kaplan–Meier, *MV* multivariate, *UV* univariate

Variables with non-significant values ($p > 0.05$) are not shown

respectively ($p = 0.002$). Similarly, in the subset of 151 patients with negative margins, 7 of 37 (18.9 %) non-irradiated patients failed locally compared with 9 of 114 (7.9 %) irradiated patients, with 3-year local control rates of 80.0 and 91.8 %, respectively (Fig. 1c; $p = 0.04$). The local control rates between patients treated with RT who had positive and negative margins were similar (85.1 and 91.8 % at 3 years, respectively; $p = 0.39$).

With respect to LRC by nodal status, patients with both pathologic and clinical node-positive disease had improved rates of LRC when treated with RT (3-year LRC rates of 75.9 and 26.7 %, $p < 0.001$; and 74.6 and 0 %, $p = 0.003$, respectively) compared with non-irradiated patients. In contrast, no difference in LRC was observed among patients with pathologic node-negative disease treated with or without RT (3-year LRC rates 83.7 and 76.0 %, $p = 0.50$, respectively).

Survival Outcomes

Twenty-seven of 41 (65.9 %) patients not treated with RT experienced a recurrence or death at last follow-up compared with 69 of 130 (53.1 %) irradiated patients. Similarly, 21 (51.2 %) patients not treated with RT had died at last follow-up compared with 57 (43.8 %) of irradiated patients. A smaller proportion of patients treated with RT died from MCC (15.4 and 22.0 %, respectively) and more were alive without evidence of disease at last follow-up (50.8 and 43.9 %, respectively) compared with patients not treated with radiation (Table 2). Patients treated with and without RT had similar rates of death from unknown or other causes (28.5 and 29.3 %, respectively). Treatment with RT was associated with both improved DFS (Appendix Fig. 1a) (70.1 and 45.6 % at 3 years, respectively; $p = 0.002$), and OS [Appendix Fig. 1b] (79.4 and 59.6 % at 3 years, respectively; $p = 0.03$) on log-rank univariate analysis. In contrast, the 3-year DSS of patients

TABLE 2 Disease status among Merkel cell carcinoma patients treated with and without radiation therapy

All patients ($n = 171$)	RT ($n = 130$)	No RT ($n = 41$)
Died of disease	20 (15.4)	9 (22.0)
Died of other/unknown causes	37 (28.5)	12 (29.3)
Alive with disease	7 (5.4)	2 (4.9)
Alive with no evidence of disease	66 (50.8)	18 (43.9)

Data are expressed as n (%)

RT radiation therapy

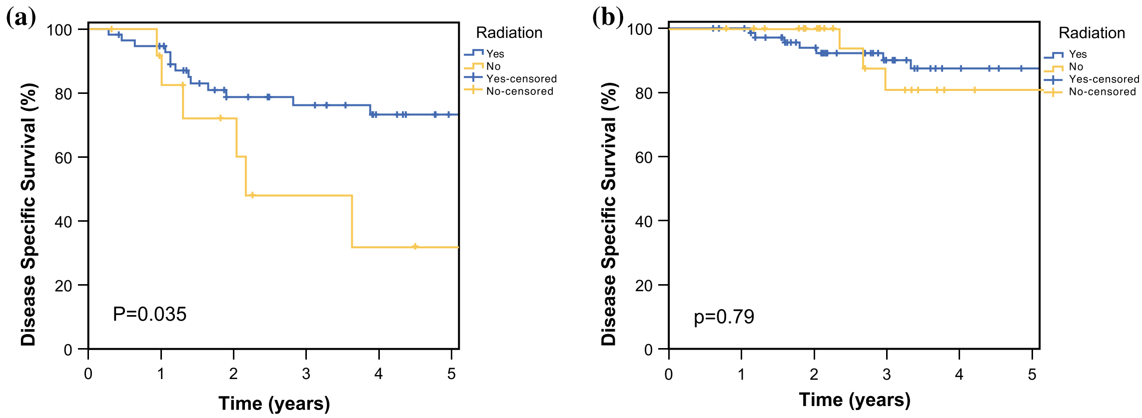

FIG. 2 Kaplan–Meier plots for patients with Merkel cell carcinoma treated with or without radiation therapy demonstrating disease-specific survival among patients with **a** node-positive disease ($n = 70$) and **b** node-negative disease ($n = 101$)

treated with and without RT was not significantly different among all patients, at 84.1 and 70.9 %, respectively (Appendix Fig. 1c) ($p = 0.23$). When assessed by lymph node status, RT was associated with improved 3-year DSS among node-positive patients [Fig. 2a] (76.2 vs. 48.1 %; $p = 0.035$) but not node-negative patients [Fig. 2b] (90.1 and 80.8 %; $p = 0.79$). On Cox multivariate analysis, RT was associated with improved survival for all three endpoints (Appendix Table 2), including DFS (HR 0.42, 95 % CI 0.26–0.70; $p = 0.001$), OS (HR 0.53, 95 % CI 0.31–0.93; $p = 0.03$), and DSS (HR 0.39, 95 % CI 0.17–0.90; $p = 0.03$).

DISCUSSION

Among 171 patients with non-metastatic MCC and pathologic evaluation of the lymph nodes, we demonstrated a higher rate of local control, LRC, and survival with the use of RT. The benefit of RT for local and LRC has previously been established in multiple retrospective series,[15–24] although a survival benefit with RT has been less pronounced in previous studies.

A recent study from Switzerland by Ghadjar et al.[20] included 180 patients with local (81 %) or regional (19 %) MCC, and assessed the utility of adjuvant RT. Similar to the present series, RT improved local relapse-free survival, regional relapse-free survival, and distant metastasis-free survival. The actuarial DFS rate with RT reported in the study (59 %) is close to that reported in the present study (70 %), and similar again to another large series reporting outcomes with combined modality therapy for MCC (60 %).[19] Ghadjar et al.[20] reported an OS rate of 56 and 46 % with and without RT, respectively, which was not significantly different ($p = 0.2$) between the radiated and non-irradiated groups. In contrast, the present study included a higher proportion of patients with lymph-node positive disease (41 and 19 %, respectively), which might account for the finding of a survival benefit with RT.

Only a single prospective trial has been reported to date assessing the utility of regional RT with relatively low-risk, stage I MCC. Jouary et al.[26] addressed the utility of local radiation alone compared with local radiation and prophylactic lymph node RT in patients with localized and clinically lymph node negative MCC after wide excision. The study included 83 patients, 44 (53 %) treated with local RT alone and 39 (47 %) treated with local plus prophylactic regional RT, and reported significantly reduced regional recurrence rates with the addition of regional RT (0 and 16.7 %, respectively; $p = 0.007$). The study supports the use of prophylactic lymph node RT in patients who are clinically lymph node-negative as assessed by physical examination and imaging, although these results are difficult to apply in the modern clinical setting, where patients typically undergo pathologic evaluation of the lymph nodes either by sentinel lymph node biopsy or lymph node dissection. The present study excluded patients with clinical node-negative disease in an attempt to represent a more modern treatment population, all who underwent pathologic nodal evaluation. As a result, findings from Jouary et al.[26] and the present study suggest that irradiation of the regional lymphatics is generally recommended for patients without pathologic lymph node evaluation, or with pathologically positive lymph nodes. Lymph node irradiation for patients with pathologically negative nodes should be individualized, but could be considered in those with high-risk features of the primary tumor, including a large tumor size (>2.0 cm), perineural or lymphovascular invasion, and/or positive margins.

In 2007, Mojica et al.[25] evaluated the benefit of adjuvant RT among 1665 patients in the SEER database with stage I (55 %), stage II (31 %), or stage III (6 %) MCC. The group reported an OS benefit with RT among all patients, along with a survival benefit among subsets of patients with tumors <1 cm ($p = 0.0447$), 1–2 cm ($p = 0.01$), and >2 cm ($p = 0.003$). These results supported the use of RT and are promising, although the SEER database lacks specific case details, including type of surgery, surgical margin status, tumor recurrence rates and sites, and RT details such as RT dose and fractionation. A subsequent study by Kim and Choi[27] included a propensity-based score analysis of a similar SEER MCC patient population treated with and without RT, and reported an OS, but not DSS, benefit with postoperative RT. Similarly, we present a lack of DSS benefit with RT on univariate analysis among all patients. However, when we looked further, we found a DSS benefit among node-positive patients but not node-negative patients treated with RT. The DSS difference was then confirmed on multivariate analysis, which also included nodal status in the final model. As such, we believe that there is likely a survival benefit among the higher-risk node-positive patients, although we cannot also exclude that the survival benefit with RT is not a result of unmeasured selection bias or factors not accounted for in the patient or treatment characteristics.

We are cautious about our recommendations regarding RT in the setting of this retrospective analysis. Because we do not have specific information about the cause of death in many patients, OS becomes the most reliable endpoint. In this context, we understand that the blanket recommendation that adjuvant RT benefits all patients is not likely to be true. On both extremes, it certainly seems that adjuvant locoregional RT strongly benefits node-positive patients, and may not benefit pathologically node-negative patients with very small primary tumors. For the bulk of the remaining patients, while our data support strong consideration of adjuvant RT, especially those with adverse prognostic factors, treatment decisions should be individualized.

CONCLUSIONS

RT for MCC was independently associated with improved local control, LRC, and survival. Local RT was beneficial regardless of margin status, while locoregional RT was beneficial for node-positive patients, but not node-negative patients. Wide excision remains the standard of care for MCC. However, given our data and other small experiences with RT alone, definitive RT appears to be a reasonable option for patients who are not surgical candidates. Future clinical initiatives could also examine topics such as the role of sentinel lymph node biopsy, the role of elective nodal RT, and other ideas aimed at the reduction of patient morbidity and financial cost of therapy.

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CONFLICT OF INTEREST Jonathan S. Zager: Consultant/Scientific Advisory Board, Delcath Systems, Inc.; Consultant/Advisory Board, IGEA. Jane L. Messina: Consultant, Glaxo Smith Kline and Durect Corporation. Vernon K. Sondak: Consultant/Advisory Board, Amgen, BMS, Merck, Navidea, Novartis, and Provectus. Javier F. Torres-Roca: Shareholder/Officer, Cvergenix, Inc., and has several awarded and pending patents for radiosensitivity gene signatures not addressed in this manuscript. Tobin Strom, Michael Carr, Arash Naghavi, Franz O. Smith, C. Wayne Cruse, Jeffery Russell, Nikhil G. Rao, William Fulp, Sungjune Kim, Tapan A. Padhya, Andy M. Trotti, Louis B. Harrison, and Jimmy J. Caudell have no conflicts of interest to declare.

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